

developed via the DevOps lifecycle are:

- Planning
- Continuous development
- Testing
- Security
- Packaging and registry
- Provisioning
- Deploying and releasing
- Operating and observing
- CI automation and cloud-native platform (central core)
- Cross-cutting dimensions

Let's now map open source tools to these phases of a DevOps lifecycle.

Plan and discover: This phase focuses on architectural decision-making, requirements gathering, issue tracking, and software catalogues. It bridges product management with platform visibility, ensuring teams can discover existing microservices, track dependencies, and manage agile workflows without fragmented visibility. Open source tools used in this phase include:

- **Backstage:** This developer portal framework unifies software catalogues, technical documentation, and developer templates into a centralised dashboard. It standardises service discovery and accelerates developer onboarding.
- **Plane/Open Project:** Plane offers AI-native tracking, while Open Project focuses on classic enterprise task management and Gantt chart planning. These tools are self-hostable for full data sovereignty and have a modern UI as well as lower resource overhead compared to legacy enterprise platforms.

Other tools that can be used are Git and Architecture Decision Records (ADRs).

Develop: This phase encompasses source code management, local environment setup, version control, and automated dependency maintenance. It focuses on reducing developer friction through automated dependency updates and standardised source code workflows. Open source tools used in

this phase are:

- **Git:** This *de facto* distributed version control system is used to track changes, handle branching, and coordinate work among developers globally.
- **Gitea/GitLab CE (Community Edition):** These self-hosted Git service platforms provide repository hosting, code review interfaces, user management, and issue tracking. They have full control over IP and source code storage. Gitea is lightweight and can run on low-resource nodes, while GitLab has feature-rich integrated pipelines.

Renovate is another tool that is used in this DevOps phase.

Test: This phase guarantees software quality, functional correctness, and performance before deployment. Modern testing relies heavily on ephemeral infrastructure (like containerised test dependencies) and automated end-to-end user path verification. Open source tools used in this phase include:

- **JUnit/pytest:** These are foundational unit testing frameworks for Java (JUnit) and Python (pytest) to validate core execution logic. They offer simple assertion structures, extensive ecosystem plugins, and native integration with all CI engine tools.
- **Playwright/Selenium:** These are automated browser testing frameworks for end-to-end (E2E) web application and cross-browser testing. They support all major modern web engines and enable headless execution in CI. Playwright offers high reliability and speed.
- **OWASP ZAP:** This open source dynamic application security testing (DAST) scanner is used to find security vulnerabilities in web applications at runtime.

Secure: This phase embeds security controls directly into the developer workflow (shift-left security). It spans

Static Application Security Testing (SAST), Software Composition Analysis (SCA), container image scanning, cryptographic signing, and software bill of materials (SBOM) generation. Open source tools that are used in this phase are:

- **Trivy:** A comprehensive security scanner for container images, file systems, Git repositories, and infrastructure as code (IaC) configurations, this tool integrates CLI and CI easily. It covers OS packages and language-specific dependencies.
- **Syft/Grype:** These tools generate detailed SBOMs from container images and file systems; Grype scans these SBOMs for vulnerability matches.

Other tools that can be used are Cosign, Snyk (Community), and Checkov.

Package and registry: This phase stores, indexes, and manages compiled software binaries, container images, Helm charts, and dependency packages in central and secure artifact repositories. Open source tools used are:

- **Harbor:** An enterprise-class open source OCI and container image registry that secures artifacts with policies and role-based access control.
- **OCI Registries/Nexus Repository (OSS):** These generic and container artifact repositories are designed to hold binaries, Maven/npm/PyPI packages, and container images.

Provision: This phase abstracts cloud and on-premises infrastructure lifecycle management into declarative code (IaC). It enforces consistent, auditable environment provisioning across public and private cloud providers. Open source tools used include:

- **OpenTofu:** This community-driven fork of Terraform (under the Linux Foundation) is used to declaratively provision cloud infrastructure.